

Task 1: Build and Deploy a Domain-Specific Chatbot

Objective

In this assignment, you will design and deploy a chatbot tailored for a specific domain. The goal is to test your understanding of model selection, pipeline design, prompt engineering, inference optimization, and deployment using Gradio.

Domain: Healthcare Information Assistant

Your chatbot should assist users in tasks such as:

- **Providing general health FAQs** (e.g., “What are the symptoms of flu?”, “How often should I exercise?”)
- **Giving information about healthy lifestyle tips** (e.g., diet, sleep, hydration)
- **Offering hospital/clinic support info** (e.g., working hours, appointment booking info, contact numbers — you can create a mock dataset)

 **Note:** The chatbot should clearly state: *“This is for educational purposes only and not a substitute for professional medical advice.”*

Tasks

1. Model Selection (10 marks)

- Choose an appropriate model (pre-trained LLM, fine-tuned model, or lightweight alternative).

- Justify why this model is suitable for the chosen domain (e.g., small model for fast responses, larger model for richer answers).

2. Pipeline Creation (15 marks)

- Design a pipeline that integrates:
 - Input processing (user queries)
 - Model inference
 - Domain-specific knowledge integration (simple FAQ dataset or knowledge base)
 - Output formatting

3. Prompt Design (15 marks)

- Write effective prompts that guide the model toward accurate and helpful responses.
- Show at least **two versions** (initial + improved) and explain the changes.

4. Inference Optimization (15 marks)

- Apply techniques such as:
 - Reducing model size (distilled models)
 - Quantization or caching frequent answers
 - Limiting response length
- Briefly explain the trade-offs made.

5. Deployment with Gradio (20 marks)

- Create a simple and user-friendly Gradio interface.
- The interface should allow a user to ask a question and receive a chatbot response.

6. Reflection & Report (25 marks)

- Write a short report (1–2 pages) covering:
 - Why you chose the model
 - How your pipeline works
 - Examples of prompts and responses
 - What optimization strategies you tried
 - Screenshots of your Gradio app
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Deliverables

- Python notebook or script with working code
 - Gradio interface (can be local, no need to host on cloud)
 - A small report in PDF/Word
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Evaluation Criteria

- Model choice & justification: **15**
 - Pipeline design: **20**
 - Prompt engineering: **10**
 - Inference optimization: **20**
 - Deployment (Gradio UI): **10**
 - Reflection & report: **25**
- Total: 100 marks**